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FullStackX: The 6-Week Accelerator

JUNE 2026

Building the Next Generation of Software, powered by AI

Course Overview

FullStackX is an immersive **6-week** program created for individuals with minimal to no programming experience who are eager to explore how software development is evolving in the age of AI. The curriculum guides participants through a journey from using AI as a creative assistant to mastering structured software engineering workflows,essential for the software stack of the future, **as being applied in Industry grade applications.**

With approximately **60% of time devoted to hands-on labs**, learners get real experience using modern AI-driven tools such as **GitHub, GitHub Copilot, GitHub Agent, Claude Code, Gemini 2.5, GLM 4.5, Bolt, Lovable, Replit, Supabase, SpecKit,** and **Vercel** for deployment.

By course end, participants will not only have built and shipped a full-stack application, but will understand how to collaborate, leverage AI, integrate databases, deploy to the cloud, and maintain code quality, essentially stepping into the world of tomorrow's "AI-enabled engineer."

Target Audience

- Absolute beginners with no coding background, who want to break into software development in an AI-rich era.
- Aspiring developers and non-technical professionals who are curious about how AI is transforming the developer workflow.
- Early programmers who are looking to integrate AI tools into their daily work and adopt modern full-stack practices.

Course Duration

- **Total Duration:** 6 Weeks
- **Hands-on Lab Sessions:** 60% of the total

Core Philosophy: "AI-First Software Engineering"

In this course we adopt a philosophy inspired by the themes inspired by Software 3.0: Software in the Age of AI is shifting, tools are evolving, and the role of the engineer is changing and is the foundation of the program.

1. **AI-Powered Creation:** For beginners, we begin with natural-language prompts and AI tools that generate functional code,starting with minimal syntax and debugging knowledge. **This phase lowers the barrier to entry and makes programming accessible.**

2. **AI-Augmented Development:** As learners progress, we introduce how AI tools (e.g., GitHub Copilot, Claude Code) assist in refining, debugging, suggesting, and accelerating code. **Developers learn to think in terms of "How can AI help me" rather than "I must write every line."**

3. **Structured Software Engineering:** Finally, we transition into disciplined engineering practices: version control, database design, specification-driven development, collaborative workflows, deployment, monitoring, and maintenance. **Learners see how AI becomes an integral part of the full software lifecycle, not just a toy.**

Together, these phases equip participants to become part of the emerging generation of engineers who build, deploy and maintain software in an AI-enhanced world.

Learning Outcomes

By the end of this course, you will be able to:

- Understand and apply the mindset of "AI-First Software Engineering" for both creation and development phases.
- Use modern AI coding tools (Claude Code, Gemini 2.5, GLM 4.5, Bolt, Lovable, Replit, GitHub Copilot, GitHub Agent) to generate, interpret, refine, and debug code.
- Employ GitHub for version control, collaborative development, branching, pull-requests, code review and merging.
- Design and interact with databases (Supabase) as part of an AI-assisted development workflow.
- Translate product requirements into actionable specification documents (PRDs) and use specification-driven tools (SpecKit) to ensure code quality and compliance.
- Architect, build and deploy full-stack applications using platforms like Vercel.
- Diagnose and troubleshoot both AI-generated and human-written code in real development contexts.
- Work effectively in collaboration with modern tools and AI assistants, in a team-like workflow.

Curriculum Outline

Week 1: Software Development Fundamentals

- **Module 1:** Git Workflows — branching, pull requests, code reviews, merging, version control in collaborative settings.

Week 2: Deployment Fundamentals

- **Module 2:** CI/CD Pipelines — continuous integration, continuous deployment, monitoring, and the basics of production software.

Week 3: Advanced GitHub & AI-Augmented Coding

- **Module 3:** Advanced GitHub & Collaborative Workflows — integrating code review, large-scale cooperation, code quality practices.
- **Module 4:** AI-Augmented Coding — using AI tools to assist writing, refactoring, debugging, accelerating development.

Week 4: Architecture Planning & Data Layer

- **Module 5:** Tools, Libraries & Artifacts — selecting frameworks, libraries, artifacts; understanding how full-stack pieces fit together.
- **Module 6:** Database Schema Design — designing data models, using Supabase, integrating with AI-enabled workflows.

Week 5: Spec-Driven Development & Capstone

- **Module 7:** Test-Driven / Spec-Driven Development with SpecKit — writing specs, verifying code against specs, ensuring code meets defined behaviors.
- **Module 8:** Capstone Project (Assigned Project, 12 Weeks Duration) — building a full-stack application from spec to deployment; applying all learned tools and practices.

Week 6: Code & Data Security

- **Module 9:** Authentication & API Security — securing applications, user auth, API gateways, permissions.
- **Module 10:** Quality Gates & Code Security — code auditing, security checks, ensuring high quality in AI-driven code environments.

Outcome

Upon completion of **FullStackX**, you will have:

- Designed, developed and deployed a full-stack application using modern AI-enabled workflows.
- Gained firsthand experience with cutting-edge AI tools in software development.
- Acquired the mindset and skill-set of the new-era software engineer: one who collaborates with AI, structures code, deploys it, maintains it, and thinks in systems,not just lines of code.

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